

Homework 0 — Mathematical Foundations

Geophys 5432: Potential Field Theory in Geophysics

Module 0: Tools, Math Review, and Course Orientation

Instructions

This problem set reviews the vector calculus and matrix algebra that underpin the rest of the course (Lecture 3). Work the problems by hand; you may verify your answers numerically in Python afterward. Show all steps — in this course the derivation matters as much as the final number.

Throughout, vectors are columns written in bold lowercase ($\mathbf{v}$) and matrices are uppercase (A). The “del” operator is $\nabla = (\partial_x, \partial_y, \partial_z)$.

Part I — Problems

Problem 1 — Vector differential operators (Cartesian)

Let the scalar field f and vector field $\mathbf{F}$ be

$$f(x, y, z) = x^2y + yz^2, \quad \mathbf{F}(x, y, z) = (xy, yz, zx).$$

- (a) Compute the gradient ∇f .
- (b) Compute the divergence $\nabla \cdot \mathbf{F}$.
- (c) Compute the curl $\nabla \times \mathbf{F}$.
- (d) Compute the Laplacian $\nabla^2 f$.

Problem 2 — Conservative, solenoidal, and finding a potential

Consider the vector field

$$\mathbf{F}(x, y, z) = (2xy + z^3, x^2, 3xz^2).$$

- (a) Show that $\mathbf{F}$ is **irrotational** ($\nabla \times \mathbf{F} = \mathbf{0}$).
- (b) Is $\mathbf{F}$ **solenoidal** ($\nabla \cdot \mathbf{F} = 0$)? Explain.
- (c) Because $\mathbf{F}$ is irrotational, a scalar potential exists. Find a function ϕ such that $\mathbf{F} = \nabla\phi$.
- (d) In geophysics the gravitational field is written $\mathbf{g} = -\nabla V$. Briefly state which of the three properties (conservative / solenoidal / irrotational) the gravity field $\mathbf{g}$ has **outside** of mass, and why.

Problem 3 — The divergence theorem

Verify the divergence theorem,

$$\int_V \nabla \cdot \mathbf{F} dV = \oint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS,$$

explicitly for the **radial field** $\mathbf{F} = (x, y, z)$ over a solid ball of radius a centered at the origin, by computing each side separately and showing they agree. Comment on how this result connects to Gauss's law for gravity.

Problem 4 — Harmonic functions and the spherical Laplacian

For a function that depends only on the radius r , the spherical Laplacian reduces to

$$\nabla^2 \phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right).$$

- (a) Show that $\phi(r) = 1/r$ is **harmonic** ($\nabla^2 \phi = 0$) for all $r > 0$. Why is this result central to gravity and magnetism?
- (b) Find the **most general** spherically symmetric harmonic function by solving $\nabla^2 \phi(r) = 0$. (Your answer should contain two arbitrary constants.)

Problem 5 — Vectors, norms, and projection (the total-field anomaly)

A total-field magnetometer does not measure the full anomalous vector field; it measures the **projection** of that vector onto the direction of the ambient main field. Work in a North–East–Down (NED) frame. Take the main-field direction to have inclination $I = 60^\circ$ and declination $D = 0^\circ$, so its unit vector is

$$\hat{\mathbf{b}} = (\cos I \cos D, \cos I \sin D, \sin I).$$

The anomalous field at a station is $\Delta\mathbf{B} = (10, 5, 20)$ nT.

- (a) Write out $\hat{\mathbf{b}}$ numerically and confirm it is a unit vector.
- (b) Compute the magnitude $\|\Delta\mathbf{B}\|$.
- (c) Compute the **total-field anomaly** $\Delta T = \Delta\mathbf{B} \cdot \hat{\mathbf{b}}$ (the scalar projection — what the instrument records).
- (d) Compute the **vector projection** of $\Delta\mathbf{B}$ onto $\hat{\mathbf{b}}$.
- (e) Explain physically why ΔT is smaller than $\|\Delta\mathbf{B}\|$.

Problem 6 — Eigenvalues, symmetry, and the condition number

Symmetric matrices appear throughout the course — as physical tensors (the gravity-gradient tensor) and as the normal-equations matrix $A^T A$ in inversion. Consider

$$M = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

- (a) Find the eigenvalues and (normalized) eigenvectors of M .
- (b) Verify the eigenvectors are orthogonal, and state the theorem that guarantees this for any real symmetric matrix.
- (c) The condition number of a symmetric positive-definite matrix is $\kappa = \lambda_{\max}/\lambda_{\min}$. Compute κ for M .
- (d) Now consider $M' = \begin{pmatrix} 1 & 0.99 \\ 0.99 & 1 \end{pmatrix}$. Find its eigenvalues and condition number, and explain in one or two sentences what a large κ implies about solving $A\mathbf{x} = \mathbf{b}$ when the data $\mathbf{b}$ contain noise. Connect this to the **non-uniqueness** problem in potential field inversion.

Part II — Solution Key

Solution 1

(a) Differentiate $f = x^2y + yz^2$ componentwise:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (2xy, x^2 + z^2, 2yz).$$

(b) With $\mathbf{F} = (xy, yz, zx)$,

$$\nabla \cdot \mathbf{F} = \frac{\partial(xy)}{\partial x} + \frac{\partial(yz)}{\partial y} + \frac{\partial(zx)}{\partial z} = y + z + x.$$

(c) Using $\nabla \times \mathbf{F} = (\partial_y F_z - \partial_z F_y, \partial_z F_x - \partial_x F_z, \partial_x F_y - \partial_y F_x)$:

$$(\nabla \times \mathbf{F})_x = \partial_y(zx) - \partial_z(yz) = 0 - y = -y,$$

$$(\nabla \times \mathbf{F})_y = \partial_z(xy) - \partial_x(zx) = 0 - z = -z,$$

$$(\nabla \times \mathbf{F})_z = \partial_x(yz) - \partial_y(xy) = 0 - x = -x.$$

So $\nabla \times \mathbf{F} = (-y, -z, -x)$.

(d) $\nabla^2 f = \partial_{xx}f + \partial_{yy}f + \partial_{zz}f$. From part (a), $\partial_{xx}f = 2y$, $\partial_{yy}f = 0$, $\partial_{zz}f = 2y$, so

$$\nabla^2 f = 2y + 0 + 2y = 4y.$$

Solution 2

(a) With $\mathbf{F} = (2xy + z^3, x^2, 3xz^2)$:

$$(\nabla \times \mathbf{F})_x = \partial_y(3xz^2) - \partial_z(x^2) = 0 - 0 = 0,$$

$$(\nabla \times \mathbf{F})_y = \partial_z(2xy + z^3) - \partial_x(3xz^2) = 3z^2 - 3z^2 = 0,$$

$$(\nabla \times \mathbf{F})_z = \partial_x(x^2) - \partial_y(2xy + z^3) = 2x - 2x = 0.$$

Hence $\nabla \times \mathbf{F} = \mathbf{0}$: the field is irrotational.

(b) $\nabla \cdot \mathbf{F} = \partial_x(2xy + z^3) + \partial_y(x^2) + \partial_z(3xz^2) = 2y + 0 + 6xz = 2y + 6xz$, which is not identically zero. So $\mathbf{F}$ is **not** solenoidal.

(c) We seek ϕ with $\nabla \phi = \mathbf{F}$. Integrate $\partial_x \phi = 2xy + z^3$ in x :

$$\phi = x^2y + xz^3 + g(y, z).$$

Then $\partial_y \phi = x^2 + g_y$ must equal $F_y = x^2$, so $g_y = 0$ and $g = h(z)$. Finally $\partial_z \phi = 3xz^2 + h'(z)$ must equal $F_z = 3xz^2$, so $h'(z) = 0$ and h is constant. Therefore

$$\boxed{\phi(x, y, z) = x^2 y + xz^3 + C.}$$

(Check: $\nabla \phi = (2xy + z^3, x^2, 3xz^2) = \mathbf{F}$.)

(d) Outside of mass, gravity is **conservative / irrotational**: $\mathbf{g} = -\nabla V$, so $\nabla \times \mathbf{g} = -\nabla \times (\nabla V) = \mathbf{0}$ automatically (curl of a gradient vanishes). It is also solenoidal *outside* the masses, since $\nabla \cdot \mathbf{g} = -4\pi G\rho = 0$ where $\rho = 0$. Inside the masses it is no longer solenoidal.

Solution 3

Left side (volume integral). For $\mathbf{F} = (x, y, z)$, $\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3$. The integrand is constant, so

$$\int_V \nabla \cdot \mathbf{F} dV = 3 \cdot \text{Vol}(V) = 3 \cdot \frac{4}{3}\pi a^3 = 4\pi a^3.$$

Right side (surface flux). On the sphere of radius a , the outward unit normal is $\hat{\mathbf{n}} = (x, y, z)/a$, so

$$\mathbf{F} \cdot \hat{\mathbf{n}} = \frac{x^2 + y^2 + z^2}{a} = \frac{a^2}{a} = a$$

(constant on the surface). Therefore

$$\oint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS = a \cdot \text{Area}(S) = a \cdot 4\pi a^2 = 4\pi a^3.$$

Both sides equal $4\pi a^3$, verifying the theorem.

Connection to Gauss's law. This is the structure of Gauss's law for gravity: the net flux of the field through a closed surface is fixed by the total “source” enclosed. For $\mathbf{g}$, $\oint_S \mathbf{g} \cdot \hat{\mathbf{n}} dS = -4\pi G M_{\text{enc}}$ — the surface flux measures the enclosed mass regardless of how that mass is distributed inside, which is one root of the non-uniqueness of potential field interpretation.

Solution 4

(a) With $\phi = 1/r$, $\frac{d\phi}{dr} = -\frac{1}{r^2}$, so $r^2 \frac{d\phi}{dr} = r^2 \left(-\frac{1}{r^2}\right) = -1$. Then

$$\nabla^2 \phi = \frac{1}{r^2} \frac{d}{dr}(-1) = 0 \quad (r > 0).$$

This matters because $1/r$ is exactly the potential of a point mass (and, with sign, a point charge / monopole). Since it is harmonic everywhere except at the source, the potential of any mass distribution is harmonic in the source-free region — which is what makes harmonic analysis (spherical harmonics, Green’s functions) the right toolkit for gravity and magnetics.

(b) Set $\nabla^2\phi = 0$:

$$\frac{1}{r^2} \frac{d}{dr}(r^2\phi') = 0 \implies \frac{d}{dr}(r^2\phi') = 0 \implies r^2\phi' = A \text{ (const).}$$

Then $\phi' = A/r^2$, and integrating gives

$$\phi(r) = -\frac{A}{r} + B.$$

Relabeling the constant ($-A \rightarrow A$), the general spherically symmetric harmonic function is

$$\phi(r) = \frac{A}{r} + B.$$

The A/r term is the point-source (external) field; the constant B is the (physically irrelevant) reference level.

Solution 5

(a) With $I = 60^\circ$, $D = 0^\circ$: $\cos 60^\circ = 0.5$, $\sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.8660$, so

$$\hat{\mathbf{b}} = (0.5, 0, 0.8660).$$

Check: $\|\hat{\mathbf{b}}\| = \sqrt{0.25 + 0 + 0.75} = \sqrt{1} = 1$.

(b) $\|\Delta\mathbf{B}\| = \sqrt{10^2 + 5^2 + 20^2} = \sqrt{100 + 25 + 400} = \sqrt{525} \approx 22.91 \text{ nT}$.

(c) Scalar projection:

$$\Delta T = \Delta\mathbf{B} \cdot \hat{\mathbf{b}} = (10)(0.5) + (5)(0) + (20)(0.8660) = 5 + 0 + 17.32 = 22.32 \text{ nT}.$$

(d) Vector projection (here $\hat{\mathbf{b}}$ is already a unit vector, so $\text{proj} = (\Delta\mathbf{B} \cdot \hat{\mathbf{b}}) \hat{\mathbf{b}}$):

$$\text{proj}_{\hat{\mathbf{b}}} \Delta\mathbf{B} = 22.32 (0.5, 0, 0.8660) = (11.16, 0, 19.33) \text{ nT}.$$

(e) The scalar projection equals $\|\Delta\mathbf{B}\| \cos \theta$, where θ is the angle between $\Delta\mathbf{B}$ and the main-field direction. Since $\Delta\mathbf{B}$ is not perfectly aligned with $\hat{\mathbf{b}}$ ($\theta \neq 0$, so $\cos \theta < 1$), the recorded total-field anomaly $\Delta T = 22.32 \text{ nT}$ is slightly less than the true vector magnitude 22.91 nT . The magnetometer “sees” only the component along the ambient field.

Solution 6

(a) Eigenvalues solve $\det(M - \lambda I) = 0$:

$$(3 - \lambda)^2 - (1)(1) = 0 \implies (3 - \lambda)^2 = 1 \implies \lambda = 4 \text{ or } \lambda = 2.$$

For $\lambda = 4$: $(M - 4I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \mathbf{v} = \mathbf{0}$ gives $v_1 = v_2$, so $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$. For $\lambda = 2$:

$(M - 2I)\mathbf{v} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0}$ gives $v_1 = -v_2$, so $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$.

(b) $\mathbf{v}_1 \cdot \mathbf{v}_2 = \frac{1}{2}[(1)(1) + (1)(-1)] = 0$, so they are orthogonal. This is guaranteed by the **spectral theorem**: every real symmetric matrix has real eigenvalues and a complete set of mutually orthogonal eigenvectors (and can be diagonalized as $M = Q\Lambda Q^T$).

(c) M is symmetric positive-definite ($\lambda_{\min} = 2 > 0$), so

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} = \frac{4}{2} = 2.$$

This is small — M is well-conditioned.

(d) For M' , the eigenvalues are $\lambda = 1 \pm 0.99$, i.e. $\lambda_{\max} = 1.99$ and $\lambda_{\min} = 0.01$, so

$$\kappa = \frac{1.99}{0.01} = 199.$$

A large condition number means the system is **ill-conditioned**: directions in model space associated with the tiny eigenvalue are barely constrained by the data, so a small amount of noise in $\mathbf{b}$ produces a large change in the solution $\mathbf{x}$. In potential field inversion this is the formal face of **non-uniqueness** — poorly constrained model parameters (small singular values) cannot be recovered reliably, which is why regularization (e.g. truncated SVD or Tikhonov) and independent geologic constraints are needed.

Numerical check (optional)

```
import numpy as np

# Problem 5
I = np.deg2rad(60); D = np.deg2rad(0)
b = np.array([np.cos(I)*np.cos(D), np.cos(I)*np.sin(D), np.sin(I)])
dB = np.array([10., 5., 20.] )
```

```

print("|dB| =", np.linalg.norm(dB))          # 22.91
print("dT  =", dB @ b)                      # 22.32
print("proj =", (dB @ b) * b)                # [11.16, 0, 19.33]

# Problem 6
M = np.array([[3., 1.], [1., 3.]])
Mp = np.array([[1., 0.99], [0.99, 1.]])
print("eig(M) =", np.linalg.eigvalsh(M))     # [2, 4]
print("kappa M =", np.linalg.cond(M))        # 2
print("eig(Mp) =", np.linalg.eigvalsh(Mp))   # [0.01, 1.99]
print("kappa Mp=", np.linalg.cond(Mp))        # 199

```